$$\hat{H} = -\frac{1}{2}\Delta_0\hat{\sigma}_x + \sum_i \left(\frac{1}{2m_i}\hat{p}_i^2 + \frac{1}{2}m_i\omega_i^2\hat{q}_i^2\right) + \hat{\sigma}_z \otimes \sum_i c_i\hat{q}_i. \tag{5.137}$$